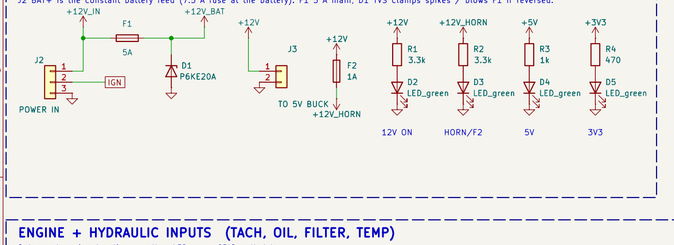
POWER IN / FUSES / PROTECTION / POWER LEDS

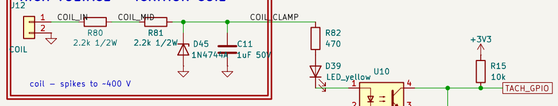
J2 BAT+ is the constant battery feed (7.5 A fuse at the battery). F1 5 A main, D1 TVS clamps spikes / blows F1 if reversed.

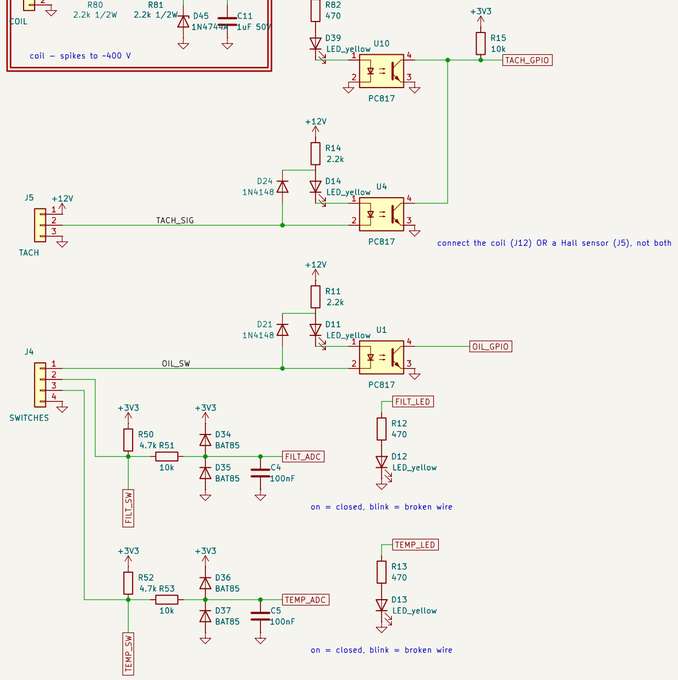


ENGINE + HYDRAULIC INPUTS (TACH, OIL, FILTER, TEMP)

Optocouplers: input active = yellow LED on = GPIO pulled low.
Filter + temp: monitored on the ADC with a 4.7k resistor AT THE SWITCH (1.65 V OK, 0 V closed, 3.3 V wire broken).

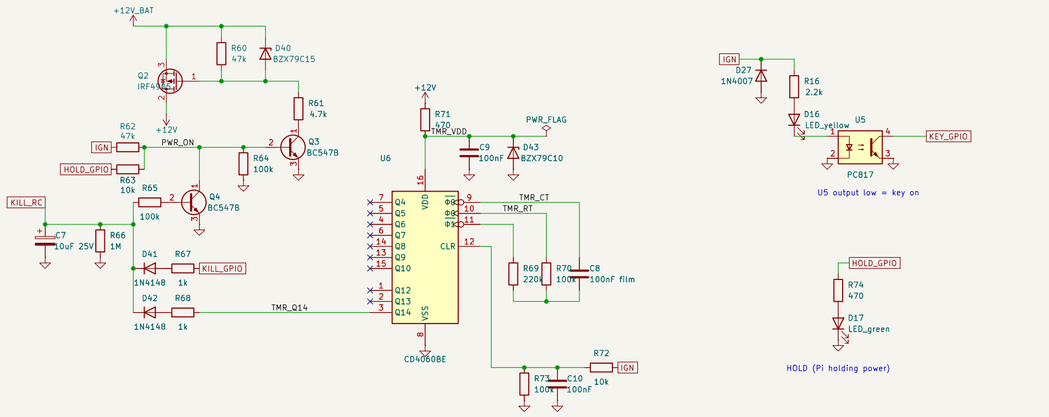
HIGH VOLTAGE – IGNITION COIL





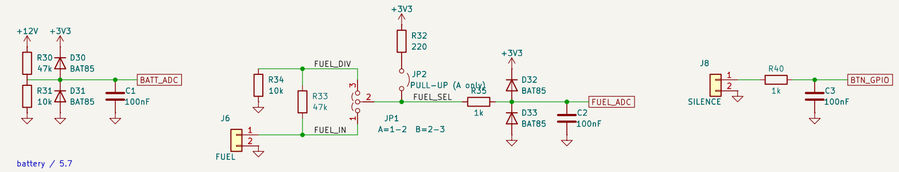
POWER HOLD

Q2 passes +12V_BAT to +12V while Q3 is on: key on (IGN) OR Pi HOLD high. Q4 cuts it: Pi KILL at the end of shutdown, or the U6 timer ~3 min after key-off. C7 stretches a cut to a few seconds.



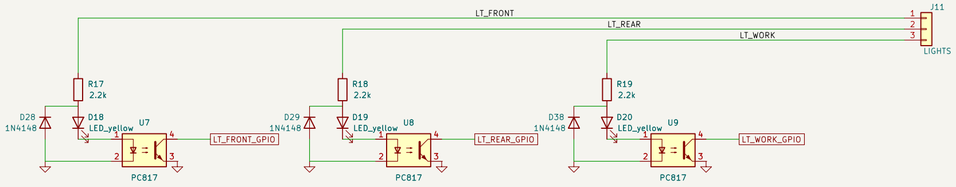
BATTERY / FUEL / BUTTON

ADC inputs clamp to 0..3.3 V. Fuel: A = sender only (JP1 1–2, JP2 on); B = gauge + dash (JP1 2–3, JP2 off).



LIGHTS ON (light sensing)

Sense wires from the LAMP side of each light switch (a few mA; the lamps' own current never comes here).

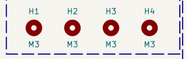


TEST POINTS

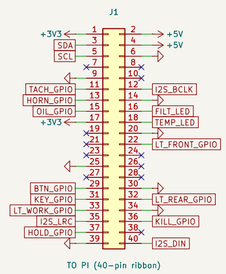
Same labels as the board silkscreen.



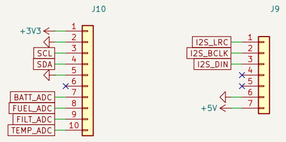
MOUNTING



TO RASPBERRY PI (40-pin ribbon, pin 1 to pin 1)



PLUG-IN MODULES

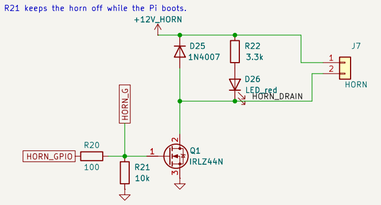


J10: ADS1115 (VDD GND SCL SDA ADDR ALRT A0–A3)
J9: MAX98357A (LRC BCLK DIN GAIN SD GND VIN)

Fuel mode: A = sender only: JP1 1–2, fit JP2. B = gauge + dash: JP1 2–3, JP2 off.
Oil + tach: +12V –> 2.2k –> yellow status LED –> PC817 LED –> sensor wire. Switch closes to chassis = LED on = GPIO low.
Filter + temp (monitored): fit a 4.7k resistor across each switch AT THE SWITCH. Wire reads ~1.65 V OK, ~0 V switch closed.
3.3 V wire broken (NO SIGNAL). Their LEDs are driven by the Pi.
F1: 5 A (Pi 4 or Pi 5). D1 (TVS) clamps spikes and blows F1 if power is connected backwards.
Power hold: +12V_IN is the CONSTANT battery feed. Q2 switches it to +12V while the key is on (IGN) or the Pi holds HOLD (GPIO26) high. At the end of shutdown the Pi raises KILL (GPIO16): Q4 cuts power. U6 cuts power ~3 min after key-off if the Pi never does. Key-off draw with everything off: under 0.1 mA.
Lights: sense wires from the LAMP side of each light switch (inline 1 A fuse at the tap). Light on = yellow LED on = GPIO low.
Tach: J12 COIL = ignition coil NEGATIVE terminal (spikes to ~400 V): R80+R81 –> 15 V zener + 1 uF –> PC817 U10.
U10 and U4 (Hall) share TACH_GPIO: connect the coil OR a Hall sensor, not both. 4-cyl coil: pulses_per_rev = 2.
Labels with the same name are connected (flag-shaped = between blocks). GND / +3V3 / +5V at the Pi header and modules are the same supplies as the ground and supply symbols.

HORN DRIVER

R21 keeps the horn off while the Pi boots.



SUPPLY FLAGS (ERC only)

Supplies enter from the harness, the buck and the Pi.



Wired schematic: one box per circuit block, the same blocks as the board silkscreen
Harness breakout between the machine wiring and the Raspberry Pi

Sheet: /
File: bobcat-dash-breakout.kicad_sch

Title: Bobcat Digital Dash – Breakout Board

Size: A2	Date: 2026-10-22	**Rev: 2.4**
KiCad E.D.A. 10.0.7		Id: 1/1